Query-Aligned Sparse STVG-R1

Initial pass: actual geometry contracts and direct versus explicit alignment

Finite scope and preserved controls

Eight fixed historical cases: four HC-STVG2 and four VidSTG. A reuses the original 48 outputs. B sees Query + paired video; C first describes marked candidates without Query, then aligns the facts to Query using the same frozen R1 in a text-only call. B / C use exactly the same eight-frame visual payload per paired request. ID swaps, panel-order swaps and mismatched-query controls are retained. No prompt retries.

P0-A: actual old 32 spatial candidates

13 / 32 original candidates contain out-of-bounds xyxy coordinates. Their original bytes / hashes are preserved; a fixed legal mapping is declared before new inference. Independent coordinate clipping, action inversion and original-z Gaussian checks pass. The sparse interface takes K <= 4 observed, verified slots. Missing receipts give K = 0. CPU forced receipts test the interface only: zero actual teacher anchors / student fits. Mapped geometry is shared by visualization, candidate readout and reward contract.

Actual output validity (exact schema, no semantic field repair)

B direct pairwise	192 / 192 valid	0 truncated
C query-free facts	118 / 144 valid	0 truncated
C text-only alignment	0 / 162 valid	0 truncated

The denominator counts actual calls; invalid upstream facts skip alignment and remain explicit missing evidence. All 162 C replies omitted only the schema field; independent raw readback found Tie / Uncertain only, no directional preference.

Do candidate-conditioned descriptions differ?

Fact field	Different A/B text	Unknown values
subject	23 / 118	14
action	11 / 118	14
relation	10 / 118	10
order	91 / 118	10
boundary	0 / 118	36
geometry	2 / 118	82

Text variation is not automatically correct localization supervision. Identity, boundary and geometry remain separate. 178 / 236 order facts say left / right. Six relation facts echo the renderer caption rather than describe a physical relation.

Query-Aligned Sparse STVG-R1

Initial pass: post-seal localization, control consistency and actual cost

Stable selections relative to mapped Frozen + Projection

Selections are sealed before GT: a unique alternative must win consistently across base / ID / order views; unsupported or conflicting evidence falls back to baseline.

Domain / branch	Method	Update	Delta pp	Pair accuracy
HC2 / temporal	B	0	+0.000	100.0% (1)
HC2 / temporal	C	0	+0.000	unknown
HC2 / spatial	B	0	+0.000	unknown
HC2 / spatial	C	0	+0.000	unknown
VidSTG / temporal	B	0	+0.000	unknown
VidSTG / temporal	C	0	+0.000	unknown
VidSTG / spatial	B	0	+0.000	unknown
VidSTG / spatial	C	0	+0.000	unknown

Accuracy excludes GT ties and nondecisive / inconsistent pairs, whose counts stay in the full anonymous table. Fallback readouts are not invented teacher choices.
A keeps its old unmodified-geometry scores; B / C are the matched new comparison.
B base pair accuracy is 75%, 58%, 64%, 75% (HC2 T/S, VidSTG T/S); ID agreement is only 9/12, 0/12, 0/12, 2/12. Base accuracy alone does not establish robust feedback.

Actual frozen-verifier cost (all positive / negative / unknown calls)

Actual calls: 498 / maximum 528; preprocess + generate: 571.25 s.
Tokens: 704,289 input / 28,059 output; peak: 16.17 GiB.
Model load: 9.64 s; no observed OOM or guard stop.
C adds text alignment cost. Frame extraction / student capture / future fits and final decode are not included in these verifier costs; this is not end-to-end TTA latency.

Decision and limitations

This fixed feedback protocol did not trigger downstream GPU adaptation.
Current saved TA-STVG output provides one trajectory, not validated different-instance hypotheses. Decoder-layer outputs are not automatically different object candidates.
A local-pool oracle bounds candidate selection, not the possible effect of parameter OPD.
Zero new student fits / backward / optimizer steps; OPD and temporal learning unmeasured.
This exposed eight-case cohort is not a fresh confirmation; reviewer GT-search incident is recorded. Raw facts, queries, RGB, annotations and model state remain private.
No new DINO / YOLO / SAM2, no carry sweep, no CURRENT promotion or old queue restart.

Query-Aligned Sparse STVG-R1

One display revision, exact C schema completion and final finite decision

Supported interface corrections; no candidate or parameter search

The initial UI / fact audit found panel-position and caption copying. One fixed display revision removed UI prose, added the same legal box in the crop and defined event-order / boundary / geometry fields. The same old-eight source cohort remained. A source audit then found an optional-version parser path and a non-JSON placeholder. Both were corrected separately with CPU contracts. No semantic response was filled. Sixteen source-role pairs were fixed: Projection vs Native; Native vs old large-plus.

Actual follow-up outputs and reused evidence

Display revision: 112 actual calls, 27 / 64 strict-JSON B replies, 0 / 48 valid facts. Exact bare categorical readback makes all 64 B replies usable as original preferences; their raw replies, prior stored parse, prompt and marked-pixel hashes remain preserved. C completion: 48 fresh facts calls; 64 exact B reuses cost zero additional inference. No complete facts record passed; therefore zero fresh text-alignment calls occurred. Failures include numeric sequence / ID content and malformed JSON, not missing data.

Keep partial facts; do not turn one bad field into a universal claim

Among 40 complete JSON packets: subject 74 / 80 valid values; action 80 / 80. Boundary: 0 candidate differences; geometry: 2 differences. Event-order values were numeric sequences, so no validated event order was obtained. Some semantic content exists. It is not a verified geometric or boundary reward. Full C protocol failure does not prove query alignment impossible or worse than B. Four short clips / paired views differ from native R1 input; this use remains unqualified.

Total finite evidence and scope of the stop

658 fresh teacher calls total (498 + 112 + 48); old A cached responses: 48. Verifier preprocess + generate: 755.69 s; peak GPU: 16.17 GiB. Tokens: 1,006,769 input; 35,755 output. No observed OOM or guard stop. Zero accepted stable improvement selections. Zero new student fits / backward / optimizer. P0-C pool expansion and P1 Direct / OPD / temporal / Combined / old-DINO control conditions were not triggered. Their effects remain unmeasured; no old queue restarts. The stop applies to this fixed feedback/input recipe only, not the whole research route. All anonymous positive / negative / neutral / unknown results and revisions are retained.